

FUNCTION

In mathematics, a function is a relation between input x and output y sets, where each x maps to a unique y , denoted as $y = f(x)$.

This notation means that for a given 'x' value, the function 'f' will produce a 'y' value.

We have conducted a study and found a linear relationship between the number of cigarettes smoked daily and the risk of developing lung cancer over a 40-year span.

- If a person does not smoke, the risk is **1%**.
- For every cigarette smoked daily, the risk increases by **0.5%**.

$$y = 0.01 + 0.005x$$

↓
Lung cancer risk
over 40 years

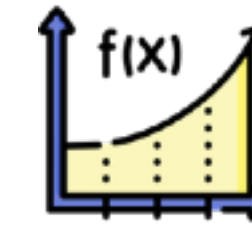
↓
The # of cigarettes
smoked daily

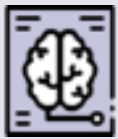
$$y = 0.01 + 0.005 (\text{8 cigarettes})$$

This individual's
40-year lung cancer
risk is 5%



FUNCTIONS IN MACHINE LEARNING



- ▶ Disease Identification from X-rays: $f(\text{X-ray image}) = \text{disease diagnosis}$ 
- ▶ Treatment Recommendations: $f(\text{patient symptoms and history}) = \text{suggested treatments}$ 
- ▶ Epidemic Outbreak Prediction: $f(\text{recent health data}) = \text{risk of epidemic outbreak}$ 
- ▶ Predictive Health Monitoring: $f(\text{real-time health metrics}) = \text{immediate health risks}$ 
- ▶ Clinical Note Analysis: $f(\text{clinical notes}) = \text{disease diagnosis or treatment recommendation}$ 
- ▶ Drug Side Effect Identification: $f(\text{patient feedback}) = \text{potential drug side-effects}$ 
- ▶ Epidemic Surveillance from News: $f(\text{news articles}) = \text{potential epidemic outbreaks}$ 
- ▶ Brain MRI Analysis: $f(\text{MRI image}) = \text{presence of tumors or anomalies}$ 
- ▶ Cell Image Classification in Pathology: $f(\text{Cell image}) = \text{cell type and potential abnormalities}$ 